

11.1 Testing reduced models

Part II fitted linear models without asking whether a fitted term was needed. Part III adds normal errors, and with them exact answers to that question. Does the murder rate of a state depend on its region once poverty and family structure are accounted for? Is a straight line enough, or is there curvature? Do two groups share a regression line? Is a coefficient equal to the value a theory predicts? Each asks whether a *reduced model*, a special case of the model already fitted, is adequate. This section sets up that comparison and derives its test. The rest of the chapter looks at the same test from other directions.

11.1.1 Hypotheses are subspaces

Throughout the chapter the data follow the normal linear model

$$\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}), \quad \boldsymbol{\beta} \in \mathbb{R}^p, \sigma^2 > 0,$$

of Eq. 7.3.1, with $r = \text{rank}(\mathbf{X}) < n$. We call it the **full model**. Its content is that the mean vector $\boldsymbol{\mu} = E(\mathbf{Y})$ lies in the r -dimensional subspace $\mathcal{C}(\mathbf{X})$ (Section 6.1). A reduced model says more: the mean lies in a smaller subspace.

Definition 11.1.1. Reduced model and linear hypothesis

Let $\mathbf{X}_0$ be an $n \times p_0$ matrix with $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ and $r_0 = \text{rank}(\mathbf{X}_0) < r$. The **reduced model** is $E(\mathbf{Y}) = \mathbf{X}_0\mathbf{c}$, $\mathbf{c} \in \mathbb{R}^{p_0}$. Inside the full model it corresponds to the **linear hypothesis**

$$H_0 : \mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0) \quad \text{against} \quad H_1 : \mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}), \mathbf{X}\boldsymbol{\beta} \notin \mathcal{C}(\mathbf{X}_0).$$

The number $q = r - r_0$ is the number of **degrees of freedom of the hypothesis**, and we write $\mathbf{M}$ and $\mathbf{M}_0$ for the orthogonal projections onto $\mathcal{C}(\mathbf{X})$ and $\mathcal{C}(\mathbf{X}_0)$.

Three features of the definition deserve comment. First, the hypothesis is a statement about the mean vector, not about coefficients. Coefficients enter only through the column spaces, so recoding the regressors of either model changes nothing (Theorem 6.7.1). Statements about coefficients, such as $\beta_2 = \beta_3$, are translated into subspaces in Section 11.3. Second, the models must be *nested*. Two models neither of which contains the other, say one with poverty and one with urbanization, are not compared by the tests of this chapter. Third, both models share the same σ^2 and the same error law. The alternative is not “anything other than the reduced model”. It is “the full model, but not the reduced one”.

The most common reduced models come from three operations on $\mathbf{X}$:

- *Deleting columns.* Dropping a block $\mathbf{X}_2$ from $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ tests whether those regressors are needed given the others. With a single column this is the familiar question about one coefficient. With all columns except the intercept it is the “overall” test of Section 11.6.
- *Merging columns.* Replacing $\mathbf{x}_2$ and $\mathbf{x}_3$ by their sum $\mathbf{x}_2 + \mathbf{x}_3$ forces equal coefficients. Merging the indicator columns of several groups forces those groups to share a mean.
- *Fixing a component.* Moving a known multiple of a column into an offset, for example testing that a slope equals 1, gives a reduced model of the form $E(\mathbf{Y}) = \mathbf{X}_0\mathbf{c} + \mathbf{X}\mathbf{b}_0$ with $\mathbf{b}_0$ known. It is handled by subtracting $\mathbf{X}\mathbf{b}_0$ from the data (Section 11.3).

11.1.2 The F statistic

If the reduced model is adequate, forcing the fit into $\mathcal{C}(\mathbf{X}_0)$ should cost little: the residual sum of squares $\text{SSE}_0 = \|(\mathbf{I} - \mathbf{M}_0)\mathbf{y}\|^2$ should exceed $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2$ by no more than noise would produce. The difference is the extra sum of squares of [Section 9.2](#), and by Eq. 6.5.2 it is a squared length,

$$\text{SSE}_0 - \text{SSE} = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_0)\mathbf{y},$$

the squared length of the projection of $\mathbf{y}$ onto the orthogonal complement of $\mathcal{C}(\mathbf{X}_0)$ within $\mathcal{C}(\mathbf{X})$. How large is “more than noise”? Noise contributes σ^2 per dimension (Theorem 9.6.1), and $\mathbf{M} - \mathbf{M}_0$ has rank q , so we compare the **hypothesis mean square** $(\text{SSE}_0 - \text{SSE})/q$ with the unbiased estimate $s^2 = \text{SSE}/(n - r)$ of σ^2 from the full model. The ratio

$$F = \frac{(\text{SSE}_0 - \text{SSE})/(r - r_0)}{\text{SSE}/(n - r)} = \frac{\mathbf{y}^\top (\mathbf{M} - \mathbf{M}_0)\mathbf{y}/q}{\mathbf{y}^\top (\mathbf{I} - \mathbf{M})\mathbf{y}/(n - r)} \quad (11.1.1)$$

is the F **statistic** for the reduced model. It is scale-free: multiplying $\mathbf{y}$ by a constant, or changing the units of the response, does not change it. Its exact distribution is the content of the first theorem of the chapter.

Theorem 11.1.1. The F test of a reduced model

Assume the normal linear model Eq. 7.3.1 and the setting of [Definition 11.1.1](#), so that $0 \leq r_0 < r < n$. Then:

- (a) $\text{SSE}_0 - \text{SSE}$ and SSE are independent, with

$$\frac{\text{SSE}_0 - \text{SSE}}{\sigma^2} \sim \chi^2(r - r_0, \gamma), \quad \frac{\text{SSE}}{\sigma^2} \sim \chi^2(n - r);$$

- (b) the statistic [Eq. 11.1.1](#) has the noncentral F distribution

$$F \sim F(r - r_0, n - r, \gamma), \quad \gamma = \frac{\|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2}{\sigma^2}; \quad (11.1.2)$$

- (c) $\gamma = \|(\mathbf{I} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2 = \min_{\mathbf{c}} \|\mathbf{X}\boldsymbol{\beta} - \mathbf{X}_0\mathbf{c}\|^2/\sigma^2$, the squared distance from the mean vector to the reduced model space in units of σ^2 . In particular $\gamma = 0$ iff H_0 holds;
- (d) under H_0 , $F \sim F(r - r_0, n - r)$, the central F distribution, whatever the values of $\boldsymbol{\beta}$ and σ^2 .

Proof. By Theorem 6.5.1, $\mathbf{M} - \mathbf{M}_0$ and $\mathbf{I} - \mathbf{M}$ are symmetric idempotent with ranks $r - r_0$ and $n - r$, and $(\mathbf{M} - \mathbf{M}_0)(\mathbf{I} - \mathbf{M}) = \mathbf{0}$. Theorem 4.4.4, applied with $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$, shows that $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}\|^2/\sigma^2$ and $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/\sigma^2$ are independent with distributions $\chi^2(r - r_0, \|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2)$ and $\chi^2(n - r, \|(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2)$. The second noncentrality is zero because $\mathbf{M}\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}$. With Eq. 6.5.2 this is (a). Part (b) follows from (a) and Definition 4.1.2, since σ^2 cancels in the ratio. (It is also Theorem 4.4.6 with $\mathbf{P} = \mathbf{M}$ and $\mathbf{P}_0 = \mathbf{M}_0$.) For (c), $\mathbf{M}\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}$ gives $(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta} = (\mathbf{I} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}$, which is the difference between $\mathbf{X}\boldsymbol{\beta}$ and its nearest point in $\mathcal{C}(\mathbf{X}_0)$ (Theorem 6.3.1). Its length is therefore the distance from $\mathbf{X}\boldsymbol{\beta}$ to $\mathcal{C}(\mathbf{X}_0)$, which is zero iff $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)$. Part (d) is (b) with $\gamma = 0$. ■

Part (d) is what makes a test possible. The reduced model is a *composite* hypothesis: it leaves $\mathbf{c}$ and σ^2 unspecified. Yet the null distribution of F is the same at every point of it. The unknown coefficients cancel because $(\mathbf{M} - \mathbf{M}_0)\mathbf{X}_0\mathbf{c} = \mathbf{0}$ and $(\mathbf{I} - \mathbf{M})\mathbf{X}_0\mathbf{c} = \mathbf{0}$, and the unknown scale cancels in the ratio. A statistic with this property is called *pivotal* under the hypothesis.

Part (c) says what the test detects. The alternative is measured by one number, the squared distance of the true mean from the reduced model, relative to the noise variance. Two alternatives with the same γ are equally hard to detect, however different their coefficients. [Section 11.5](#) is built on this.

11.1.3 The test and its p-value

Large values of F are evidence against H_0 : under the alternative the numerator mean square estimates $\sigma^2 + \sigma^2\gamma/q$ while the denominator still estimates σ^2 (Eq. 9.6.1). The F test at level α rejects H_0 when

$$F > F_\alpha(r - r_0, n - r),$$

where $F_\alpha(a, b)$ is the upper α point of the central $F(a, b)$ distribution. By [Theorem 11.1.1\(d\)](#) its probability of rejecting is exactly α at every point of the reduced model. It rejects with probability greater than α at every point of the alternative, by (c) and Theorem 4.1.4. Instead of a fixed level, one usually reports the **p-value**

$$p(\mathbf{y}) = \Pr\{F(r - r_0, n - r) \geq F_{\text{obs}}\},$$

the probability under the reduced model of a statistic at least as extreme as the one observed.

Proposition 11.1.2. The p-value is uniform under the hypothesis

Under H_0 , the p-value $p(\mathbf{Y})$ has the uniform distribution on $(0, 1)$. For every $\alpha \in (0, 1)$, rejecting when $p(\mathbf{Y}) \leq \alpha$ is the F test at level α .

Proof. Let Φ_F be the distribution function of $F(q, n - r)$. It is continuous and strictly increasing on $(0, \infty)$, since the distribution has a positive density there, and $p(\mathbf{y}) = 1 - \Phi_F(F_{\text{obs}})$. Under H_0 the statistic F has distribution function Φ_F , so for $u \in (0, 1)$, $\Pr\{p(\mathbf{Y}) \leq u\} = \Pr\{\Phi_F(F) \geq 1 - u\} = \Pr\{F \geq \Phi_F^{-1}(1 - u)\} = u$. Finally, $p(\mathbf{y}) \leq \alpha$ iff $\Phi_F(F_{\text{obs}}) \geq 1 - \alpha$ iff $F_{\text{obs}} \geq F_\alpha(q, n - r)$, and the event $F = F_\alpha$ has probability zero. ■

The p-value is a statement about the data under H_0 . It is *not* the probability that H_0 is true, and it is not the probability that the result is “due to chance”. A small p-value says that data like these would be unusual if the reduced model held.

Example. Do regions matter?

In the 2009 data on 50 US states used in [Chapter 6](#) and [Chapter 9](#), regress the murder rate on the poverty rate, the percentage of single-parent households and the percentage living in urbanized areas, with an intercept. This is the reduced model, with $r_0 = 4$. The full model adds indicators of three of the four Census regions (Northeast, South, West, with the Midwest as reference), so $r = 7$. The question is whether the regions differ in murder rate once the three covariates are accounted for. The listing gives

Source	Degrees of freedom	Sum of squares	Mean square
Regions, given the covariates	3	14.83	4.944
Residual, full model	43	87.13	2.026
Residual, reduced model	46	101.96	

so $F = 4.944/2.026 = 2.440$ on 3 and 43 degrees of freedom. The 5% critical value is 2.822 and the p-value is 0.077. At the conventional level the reduced model is not rejected. The data do not show convincingly that the regions differ, given the covariates.

Yet in the full model the Northeast coefficient is -1.736 murders per 100,000, with a t statistic of -2.519 and a p-value of 0.016. One of the three contrasts looks clearly nonzero while the joint test does not reject. There is no contradiction: [Section 11.6](#) shows how a joint test can dilute one strong effect among weak ones, and [Chapter 13](#) explains why a contrast singled out after looking at the estimates cannot be judged by its own p-value.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statelcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
# Census region of each state, in the data's alphabetical order (N, M, S, W)
codes = "SWWSWNSSSWMMMMSSNSNMMSMWNNWNSMMSWNNSSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
covariates = np.column_stack([data["poverty"], data["single"], data["urban"]])
X0 = np.column_stack([np.ones(n), covariates]) # reduced model
D = np.column_stack([(region == g).astype(float) for g in "NSW"]) # Midwest is the
# reference
X = np.column_stack([X0, D]) # full model

def sse(Z):
    """Residual sum of squares of the least squares fit of y on the columns of Z."""
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    return np.sum((y - Z @ b) ** 2)

r, r0 = np.linalg.matrix_rank(X), np.linalg.matrix_rank(X0)
sse_full, sse_reduced = sse(X), sse(X0)
F = ((sse_reduced - sse_full) / (r - r0)) / (sse_full / (n - r))
p_value = stats.f.sf(F, r - r0, n - r)
crit = stats.f.ppf(0.95, r - r0, n - r)
print(f"SSE0 = {sse_reduced:.2f}, SSE = {sse_full:.2f}, df = ({r - r0}, {n - r})")
print(f"F = {F:.3f}, p = {p_value:.4f}, 5% critical value = {crit:.3f}")
```

[Run this code online](#)

11.1.4 What the test assumes

[Theorem 11.1.1](#) rests on three assumptions.

The full model must hold. The denominator s^2 estimates σ^2 only if $E(\mathbf{Y}) \in \mathcal{C}(\mathbf{X})$. If the

full model omits something, $E(s^2)$ exceeds σ^2 by $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2/(n - r)$ (Theorem 9.6.1(b)), the denominator is inflated, and F is pushed towards zero. The test then loses power, and a nonsignificant result says even less than usual. The test of a reduced model is a test *within* the full model. It cannot certify the full model itself. Chapter 14 tests the full model against a nonparametric alternative when there are replicate observations.

The errors must be independent with a common variance. Correlated or heteroscedastic errors change both numerator and denominator, and the null distribution is no longer F . Chapter 21 and Chapter 31 treat these cases.

Normality gives the exact distribution, and it is the least critical of the three. The null distribution is exactly F for every spherically symmetric error law (Section 11.2), and approximately so for independent nonnormal errors unless the sample is small or some points have high leverage (Theorem 19.3.2).

Warning (What a nonsignificant F does not show). Failing to reject the reduced model is *not* evidence that the reduced model is true. The test controls the probability of wrongly rejecting H_0 . It says nothing on its own about the probability of wrongly accepting it, which depends on the unknown noncentrality γ and may be large. In Example (Do regions matter?), $p = 0.077$ is compatible with no regional differences. It is equally compatible with sizeable regional differences. Section 11.5 computes that the test had only about an even chance of detecting a Northeast shortfall of 1.5 murders per 100,000, about the size estimated in Section 11.3 (1.379 below the other three regions). To support “the reduced model is adequate”, one must show that every alternative of practical importance would have been rejected. That calls for a power calculation made before the data were seen, or better, a confidence region for the omitted effects that excludes all important values (Chapter 12), or a test of equivalence in which the roles of hypothesis and alternative are reversed. Nor does a nonsignificant F license dropping the terms and then reporting the reduced fit as if it had been specified in advance.

Warning (What a significant F does not show). A significant F says that the mean vector is not in $\mathcal{C}(\mathbf{X}_0)$, assuming the full model holds. It does not say which of the q dimensions of the test space carries the departure, nor that every omitted coefficient is nonzero. It does not say that the departure is large enough to matter: with large n , a tiny γ/n is detected with certainty (Section 11.5). And it does not say that the full model is correct. The statistic only compares two models, and a third, better model may exist.

11.1.5 Exercises

11.1.5.1 A. Check your understanding

Exercise. A1

Show that the F statistic Eq. 11.1.1 does not change when $\mathbf{y}$ is replaced by $a\mathbf{y} + \mathbf{X}_0\mathbf{c}$ for any $a \neq 0$ and any vector $\mathbf{c}$. Interpret the result for the region test of Example (Do regions matter?) when murder rates are expressed per million instead of per 100,000.

Exercise. A2

In the model $E(Y_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3}$, write down a matrix $\mathbf{X}_0$ (or an offset and an $\mathbf{X}_0$) for each reduced model and give q : (i) $\beta_2 = \beta_3 = 0$; (ii) $\beta_1 = \beta_2 = \beta_3$; (iii) $\beta_1 = 2$; (iv) $\beta_2 + \beta_3 = 0$.

11.1.5.2 B. Practice**Exercise. B1**

Suppose $\sigma^2 = \sigma_0^2$ is known. Show that $X^2 = (\text{SSE}_0 - \text{SSE})/\sigma_0^2$ has the $\chi^2(q, \gamma)$ distribution and that the test rejecting when X^2 exceeds the upper α point of $\chi^2(q)$ has level α . Show that $qF/X^2 \rightarrow 1$ in probability as $n \rightarrow \infty$ with q fixed and the full model true, and explain why the F test and this test then behave alike.

Solution. The distribution is [Theorem 11.1.1\(a\)](#) with $\sigma^2 = \sigma_0^2$; under H_0 it is $\chi^2(q)$, so the level is exact. Now $qF/X^2 = \sigma_0^2/s^2$, and $s^2 \rightarrow \sigma^2 = \sigma_0^2$ in probability, since $E(s^2) = \sigma^2$ and $\text{Var}(s^2) = 2\sigma^4/(n-r) \rightarrow 0$ ([Theorem 7.5.1\(d\)](#)) and Chebyshev's inequality applies. So qF and X^2 differ by a factor tending to one, and $qF_\alpha(q, n-r)$ tends to the upper α point of $\chi^2(q)$ for the same reason. With few residual degrees of freedom the two tests differ, and only the F test has the right level when σ^2 must be estimated.

Exercise. B2

Suppose the full model is wrong: $E(\mathbf{Y}) = \boldsymbol{\mu}$ with $(\mathbf{I} - \mathbf{M})\boldsymbol{\mu} = \boldsymbol{\delta} \neq \mathbf{0}$, but $(\mathbf{M} - \mathbf{M}_0)\boldsymbol{\mu} = \mathbf{0}$. Show that the numerator and denominator of [Eq. 11.1.1](#) are still independent, that $E(\text{numerator}) = \sigma^2$ and $E(\text{denominator}) = \sigma^2 + \|\boldsymbol{\delta}\|^2/(n-r)$, and that $\Pr(F \leq c) \geq \Pr\{F(q, n-r) \leq c\}$ for every $c > 0$. Explain why an F statistic far below one, with a p-value near one, is a hint that the full model is inadequate.

Solution. Independence and the distributions follow from [Theorem 4.4.4](#) as in the proof of [Theorem 11.1.1](#): the numerator is $\sigma^2\chi^2(q)/q$, and $\text{SSE}/\sigma^2 \sim \chi^2(n-r, \|\boldsymbol{\delta}\|^2/\sigma^2)$, with mean $n-r + \|\boldsymbol{\delta}\|^2/\sigma^2$ ([Theorem 4.1.2\(b\)](#)). For the probability, condition on the numerator U : $F \leq c$ iff $\text{SSE}/\sigma^2 \geq (n-r)U/(qc\sigma^2)$, and a noncentral χ^2 exceeds any fixed value with at least the central probability ([Proposition 4.1.3](#)). Averaging over U gives the inequality. So omitted structure in the residual space makes small values of F more likely. Under the full model, $\Pr(F \leq c)$ is small for small c , and an observed F in the far lower tail is surprising.

Exercise. B3

Let $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_1) \subseteq \mathcal{C}(\mathbf{X})$ with ranks $r_0 < r_1 < r < n$, and let $s^2 = \text{SSE}/(n-r)$ come from the largest model. Show that

$$F_1 = \frac{\mathbf{y}^\top (\mathbf{M}_1 - \mathbf{M}_0) \mathbf{y} / (r_1 - r_0)}{s^2}$$

has the $F(r_1 - r_0, n - r, \gamma_1)$ distribution with $\gamma_1 = \|(\mathbf{M}_1 - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2$ whenever the largest

model holds, and that its numerator is independent of that of the test of $\mathbf{X}_1$ within $\mathbf{X}$. Why is this preferable to using the residual mean square of $\mathbf{X}_1$?

Exercise. B4

Two samples of sizes n_1 and n_2 follow $E(\mathbf{Y}_k) = \mathbf{X}_k \boldsymbol{\beta}_k$, $k = 1, 2$, with a common σ^2 and $\mathbf{X}_k$ of full column rank p . Write the model for both samples together as a full model with block-diagonal model matrix, and the hypothesis $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_2$ as a reduced model. Show that

$$F = \frac{(\text{SSE}_{\text{pooled}} - \text{SSE}_1 - \text{SSE}_2)/p}{(\text{SSE}_1 + \text{SSE}_2)/(n_1 + n_2 - 2p)},$$

where SSE_k comes from fitting sample k alone and $\text{SSE}_{\text{pooled}}$ from one fit to all the data. Apply the test to the question whether two firms of the Grunfeld panel (`sm.datasets.grunfeld`) share their investment equation, and comment on the assumption of a common σ^2 .

11.1.5.3 C. Going deeper

Exercise. C1

In [Exercise \(B4\)](#) suppose instead that $n_2 \leq p$ and $\text{rank}(\mathbf{X}_2) = n_2$, so that $\boldsymbol{\beta}_2$ cannot be estimated from the second sample. Show that the full model then fits the second sample exactly, identify the test space, and show that the F statistic of [Theorem 11.1.1](#) becomes

$$F = \frac{(\text{SSE}_{\text{pooled}} - \text{SSE}_1)/n_2}{\text{SSE}_1/(n_1 - p)},$$

a test of whether the second sample is predicted by the fit to the first, on n_2 and $n_1 - p$ degrees of freedom.